2019a)), but is a decision we make for practical reasons (e.g. availability of datasets, effort to implement scenarios, and computational resources to evaluate LMs on chosen scenarios).

Tasks. To select tasks, we begin with the set we described previously. Since we are studying English language models, we filter infeasible tasks (e.g. multimodal tasks or machine translation are not suitable for unimodal English language models).[33] Of the remaining tasks, we elect to prioritize *user-facing* tasks: we believe these tasks will confer much of the *direct* social impact of language models and aligns with our perspective of language models as *interfaces*. Consequently, we filter tasks based on our judgments of what is user-facing.[34] This yields the following tasks: *question answering*, *information retrieval*, *summarization*, *sentiment analysis*, and *toxicity detection*.[35] And to provide some coverage of the long tail of tasks, we include *miscellaneous text classification*, which represents the non-standard text classification use cases for language technologies historically and at present for language models.

Domains and Languages. Given that we found it more complicated to arrive at an explicit enumeration of domains compared to tasks,[36] we instead focus on domain coverage during our selection of specific datasets to instantiate scenarios. Similarly, we ensure coverage of the English varieties of different English-speaking countries as well as African American English through targeted evaluations that we discuss in §5.1: LANGUAGE. In doing so, we also demonstrate our desire for a minimal set of evaluations (both because evaluations have costs, so larger sets will be more unwieldy, and producing more results often comes at the cost of clarity on how to sift through them). With this in mind, we emphasize that for large regions of the scenario space, specifically in relation to domains (e.g. scenarios involving text written by elderly speakers), there exist very few, if any, datasets in NLP. We hope the community can build on our work by ensuring greater coverage of the domains and scenarios we did not cover in our benchmark by building the necessary and oft-undervalued resources (Jo & Gebru, 2020; Paullada et al., 2021; Rogers, 2021; Jernite et al., 2022). To facilitate this, we explicitly identify specific scenarios that we recommend prioritizing in §10.1: MISSING-SCENARIOS. We also note that there is more to a dataset than just these axes, which determine how well it operationalizes the desired use case (e.g. the quality of crowd-sourced labels in the dataset). Having settled on the tasks we will cover and our approach to domain/language coverage, we detail how we selected the particular datasets for each scenario.

3.3 Question answering

Question answering (QA) is a fundamental task in NLP that underpins many real-world applications including web search, chatbots, and personal assistants. QA is very broad in terms of the questions that can be asked and the skills that are required to arrive at the answer, covering general language understanding (§5.1: LANGUAGE), integration of knowledge (§5.2: KNOWLEDGE), and reasoning (§5.3: REASONING) (Gardner et al., 2019; Rogers et al., 2021).

Problem setting. In QA, given a question (e.g. "Where was the painter of the Mona Lisa born?"), the task is to predict the correct answer ("Italy"). The format of question answering may have some variations: in the *open-book* or *reading comprehension* setting, additional context to refer to, such as supporting documents

[33] We do note that various works have shown these models can achieve nontrivial performance on multimodal tasks (with modalities beyond text) and on other languages (especially as some of these models, most notably BLOOM (176B), GLM (130B), and YaLM (100B) are trained on sizable datasets in other languages). With that said, we expect that multimodal or multilingual approaches would be more appropriate to achieve reasonable performance for these tasks compared to these models, so we defer such evaluation to future work.

[34] We emphasize that this *does not* mean we believe the other tasks are less important nor that they should not be evaluated for in future work.

[35] We note that our interpretation of what is user-facing namely excludes tasks that are generally not the subject of applications (e.g. natural language inference) as well as many classical NLP tasks that served as intermediaries (Jurafsky & Martin, 2000) in traditional NLP pipelines (e.g. named entity recognition, part-of-speech tagging, syntactic parsing, information extraction). We also do not study interactive tasks such as dialogue, which will be discuss in forthcoming companion work associated with this effort (Lee et al., Forthcoming).

[36] Though we note this could be attempted by proposing a taxonomy for each of the 3 W's we consider and then taking the resulting Cartesian product.

Scenario: MMLU(subject=anatomy)

Input: *Which of the following terms describes the body's ability to maintain its normal state?*

References:
- *Anabolism*
- *Catabolism*
- *Tolerance*
- *Homeostasis* [correct]

Figure 11: **Example of question answering.** An example instance for question answering from MMLU. Different QA scenarios can have significantly different properties, but this example captures the overall structure of question answering.

(e.g. Wikipedia page of "Mona Lisa"), is given to the model. In the *multiple-choice* setting, answer choices to choose from (e.g. "(A) France (B) Italy") are given to the question. Figure 11 depicts an example.

Datasets and selection process. There are hundreds of question-answering datasets available in NLP, with a rapid increase in the number of datasets in recent years (Rogers et al., 2021). To select question-answering datasets, we prioritized (i) domain coverage, in terms of the domain of the inputs/contexts and (ii) coverage of component skills required for the datasets (e.g. we deliberately ensured of datasets that required commonsense knowledge and reasoning).

We selected the **NaturalQuestions** (Kwiatkowski et al., 2019), **NarrativeQA** (Kočisky et al., 2017), and **QuAC** (Choi et al., 2018) datasets to ensure domain coverage as these datasets cover web search queries, stories, and conversational questions (i.e. dialogue) respectively. **NaturalQuestions** consists of questions from queries to Google search and annotations from Wikipedia; we consider both *open-book* and *closed-book* variants of **NaturalQuestions**. **NarrativeQA** tests reading comprehension through the understanding of books and movie scripts. **QuAC** (Question Answering in Context) provides freeform questions and answers which are more open-ended and dependent on context.

To these, we add the **HellaSwag** (Zellers et al., 2019), **OpenBookQA** (Mihaylov et al., 2018), and **TruthfulQA** (Lin et al., 2021b) datasets to ensure coverage of commonsense knowledge and reasoning. **HellaSwag** tests commonsense inference and was created through adversarial filtering to synthesize wrong answers. **OpenBookQA** is based on open book exams, with a collection of basic science facts and crowd-sourced multiple-choice questions to test understanding and application of these facts. **TruthfulQA** tests model truthfulness through questions that align with common human misconceptions, spanning law, medicine, finance, and politics, among others, that were adversarially generated using davinci (175B) as the target model.

To further ensure broad coverage of knowledge-intensive question answering across many disciplines, we add the **MMLU** (Hendrycks et al., 2021c) meta-benchmark of 57 constituent datasets. **MMLU** (Measuring Massive Multitask Language Understanding) measures multitask accuracy and includes a diverse set of 57 tasks, testing problem solving and general knowledge.

Finally, we add **BoolQ** (Clark et al., 2019) which, in addition to **QuAC**, was used to study model robustness to equivariances due to the available contrast set (Gardner et al., 2020). **BoolQ** is a collection of binary yes/no questions generated through the same process as **NaturalQuestions**.

3.4 Information retrieval

Information retrieval (IR), which refers to the class of tasks concerned with searching large unstructured collections (often *text* collections), is central to numerous user-facing applications. IR has a long tradition of study (Salton & Lesk, 1965; Salton, 1971; Spärck Jones, 1972; Salton & McGill, 1983; Manning et al., 2008; Lin et al., 2021a) and is one of the most widely deployed language technologies. It powers the Web and e-commerce search, and serves as a key component in many knowledge-intensive NLP systems for open-domain question answering or fact checking.

Scenario: MS MARCO

Input: *how much does a spectacled bear weigh*

References:
- *Male spectacled bears … weigh from 120 to 340 pounds…* [rank=1]
- Spectacled Bear Description. Spectacled Bears are generally smaller … [rank=2]
- *The panda's closest relative is the spectacled bear …* [rank=3]
- …

Figure 12: **Example of information retrieval (passage ranking).** An example instance for information retrieval from MS MARCO.

We focus here on the passage ranking task: given a query q and a large corpus C of passages, systems must output a list of the top-k passages from C in decreasing "relevance" to q. We specifically study this in the context of *re-ranking*: since C is typically extremely large (e.g. $|C| > 10M$ passages), we consider only ranking the top-k passages among a set retrieved for q (i.e. $M(q)$ where $|M(q)| \ll |C|$) by an efficient external retrieval mechanism (e.g. BM25; Robertson & Zaragoza, 2009).

IR differs fundamentally from the other tasks we consider in this work, as each test example (i.e. a query) entails processing a large set of passages and will likely invoke the LM numerous times to do so.[37] Because of this, IR tasks have received very little attention in the few-shot in-context learning with language models, with the exception of the recent zero-shot approach by Sachan et al. (2022).

Problem setting. We address the re-ranking task in a *pointwise* fashion: we formulate the information retrieval problem using prompting as a binary log-probability problem, similar to Nogueira & Cho (2019): Given a passage c_i and a query q, we ask the model whether the passage contains an answer to the query. If the model's answer is `Yes` with a high probability, we rank the corresponding c_i higher, while the `No` answer with high probability achieves the opposite. Figure 12 depicts an example instance. The rankings produced are then evaluated using standard information retrieval metrics.

Datasets and selection process. We demonstrate the information retrieval task using the MS MARCO ranking datasets. While it is originally a question answering task, the retrieval version of MS MARCO is the largest publicly available collection of relevance judgments and has been central to much of the progress in neural IR over the past several years (Lin et al., 2021a).

We use the original passage ranking dataset accompanying the public MS MARCO leaderboard[38] (Nguyen et al. 2016; henceforth, the **regular** track) and the dataset from the TREC 2019 Deep Learning track (Craswell et al. 2020; henceforth, the **TREC** track). Both datasets evaluate retrieval out of a collection of 9M passages from the Web. The regular track contains a large number of queries (e.g., over 500,000 training set queries) with *sparse* relevance judgments: on average, annotators identify only one "positive" (relevant) passage for each query, and every other passage is assumed to be a negative. In contrast to this, the TREC track contains only 43 queries that are more rigorously annotated, with over 9,000 query–passage pairs with associated relevance judgments corresponding to the 43 queries.

3.5 Summarization

Text summarization is an established research direction in NLP (Luhn, 1958; Mani, 1999; Spärck Jones, 1999; Nenkova & McKeown, 2012), with growing practical importance given the ever-increasing volume of text that would benefit from summarization. To effectively summarize, systems must identify and yield the core relevant and informative content in the source document while removing less critical information and avoiding redundancy (Peyrard, 2019). The rise of language models in recent years has dramatically

[37] Effectively, this means that model outputs come from a very large combinatorial space: they are much more constrained than open-ended generation tasks but much less so than standard classification tasks, setting this scenario apart from many others in terms of its automatic evaluation.

[38] `https://microsoft.github.io/msmarco/`